Rohan Garg

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Education

Stanford University Sept 2024 – June 2026
MS in Mechanical Engineering (Robotics Specialization)
Teaching Assistant: CS238, CS234, AA212, CS237A

Indian Institute of Technology, Delhi 2019 – 2023
Mechanical Engineering, Department Rank 7/91, Best Undergraduate Thesis

Industry Experience

Field AI *Autonomy Engineer Intern | Irvine, CA* June 2025 – September 2025
- Designed and improved the reliability of the autonomy stack in ROS, C++ for Last-Hundred Yard delivery on quadrupeds
- Developed learning-based motion planning pipeline integrating human behavior prediction for socially compliant navigation

Goldman Sachs *Quantitative Analyst Summer Intern | Bangalore, India* June 2022 – August 2022
- Automated Asset Management's quarterly budgeting reducing workflow time by 20% and saved 40 analyst hours weekly.

Publications

GGraph: A GPU-Accelerated Library for Graph-Based Robot Deployment and Learning [PDF]
Yash Rampuria*, **Rohan Garg***, Daniele Gammelli, Marco Pavone **Under Review, ICRA 2027**

Research Experience

Navigation & Loco-Manipulation for Legged Robots *Prof. Marco Pavone* Dec 2025 – Present
- Designed legged-robot autonomy for semantic exploration in the Stanford–Berkeley DARPA submission, winning 2nd prize.
- Developed GGraph to build and maintain GPU-native navigation graphs from local terrain maps at scales of millions of nodes and edges.
- Enabled GNN-based exploration policies trained with RL that outperform state-of-the-art exploration planners by 20%.
- Developing agentic and RL loco-manipulation policies for legged robots to identify and move semantically appropriate objects.

Dynamics conditioned Hamilton Jacobi Reachability for drones *Prof. Somil Bansal* Dec 2024 – Aug 2025
- Derived dynamics-conditioned Hamilton-Jacobi unsafe sets for drones under wind disturbances using learning-based controls.
- Generated quadcopter trajectories in RotorPy; learnt control-affine dynamics model with wind encoded as a latent state.
- Computed Hamilton-Jacobi BRTs up to 8D; validated safety filter in ROS2 on CrazyFlie 2.1 under 2 m/s wind speed.

Battery Management System for Indian climate *Prof. Amit Gupta | IIT Delhi* 2022 – 2023
- Developed low-cost BMS for li-ion cells; using adaptive non-linear Unscented Kalman Filter for real-time SoC estimation.
- Modelled battery with temperature-dependent 2-RC ECM; estimated data at varying temperatures using battery cyclers
- Won Man Mohan Suri Award for Best UG Thesis (Mech. Dept.); invention filed as Indian Patent No. 202311067119.

Selected Projects

TidyBot++ Collaborative Robotic Assistant *CS339R (Collaborative Robotics) | [Code] [Website]* Winter 2025
- Built full autonomy stack on ROS2 for a bimanual mobile manipulator: SAM3 for 3D object detection and grasp pose estimation, and a Gemini LLM with function-calling for task planning from natural language and voice commands.
- Deployed on real hardware demonstrating object retrieval, multi-step sequential tasks, and dexterous drawer opening.

RL based Data Selection for Finetuning LLMs *CS 224R (Deep RL) | [Paper]* Spring 2025
- Formulated data selection as a Sequential Decision Making problem, learnt policies via PPO to select training subsets
- Achieved accuracy close to full-dataset training while selecting only 5% of the original data, reducing training time by 2×

Achievements

- Winner NASA Edge Autonomy Challenge at RoboHacks, a robotics hackathon hosted by Y Combinator (2026)
- 2nd place in DARPA-Transfer Abstract Models to Autonomous Tech. challenge winning total prize money of $750k (2026)
- Man Mohan Suri Award for Best Undergraduate Thesis, Mechanical Engineering Department, IIT Delhi (2023)
- Awarded the National Talent Search Examination Scholarship by the Government of India (2017)